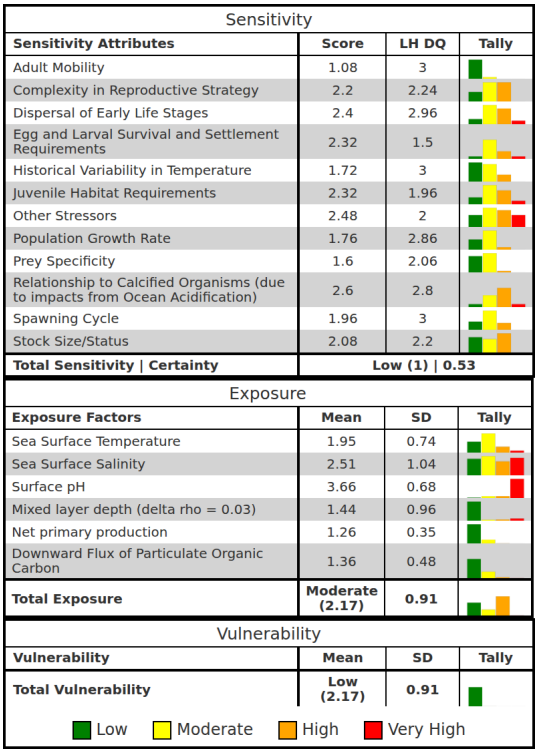
Cobia (*Rachycentron canadum*)

Summary Scores

- **Sensitivity:** Low
- **Exposure:** Moderate
- **Vulnerability:** Low



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.08	3	
Complexity in Reproductive Strategy	2.2	2.24	
Dispersal of Early Life Stages	2.4	2.96	
Egg and Larval Survival and Settlement Requirements	2.32	1.5	
Historical Variability in Temperature	1.72	3	
Juvenile Habitat Requirements	2.32	1.96	
Other Stressors	2.48	2	
Population Growth Rate	1.76	2.86	
Prey Specificity	1.6	2.06	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	2.6	2.8	
Spawning Cycle	1.96	3	
Stock Size/Status	2.08	2.2	
Total Sensitivity | Certainty	**Low (1) | 0.53**		

Exposure

Exposure Factors	Mean	SD	Tally
Sea Surface Temperature	1.95	0.74	
Sea Surface Salinity	2.51	1.04	
Surface pH	3.66	0.68	
Mixed layer depth (delta rho = 0.03)	1.44	0.96	
Net primary production	1.26	0.35	
Downward Flux of Particulate Organic Carbon	1.36	0.48	
Total Exposure	**Moderate (2.17)**	**0.91**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (2.17)**	**0.91**	

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Cobia. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Moderate (Burton et al. 2023)	Low	Decrease
Exposure	Very High (Burton et al. 2023)	Moderate	Decrease
Vulnerability	High (Burton et al. 2023)	Low	Decrease
Directional Effect	Neutral (Burton et al. 2023)	Neutral	Decrease

Stocks

Cobia is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

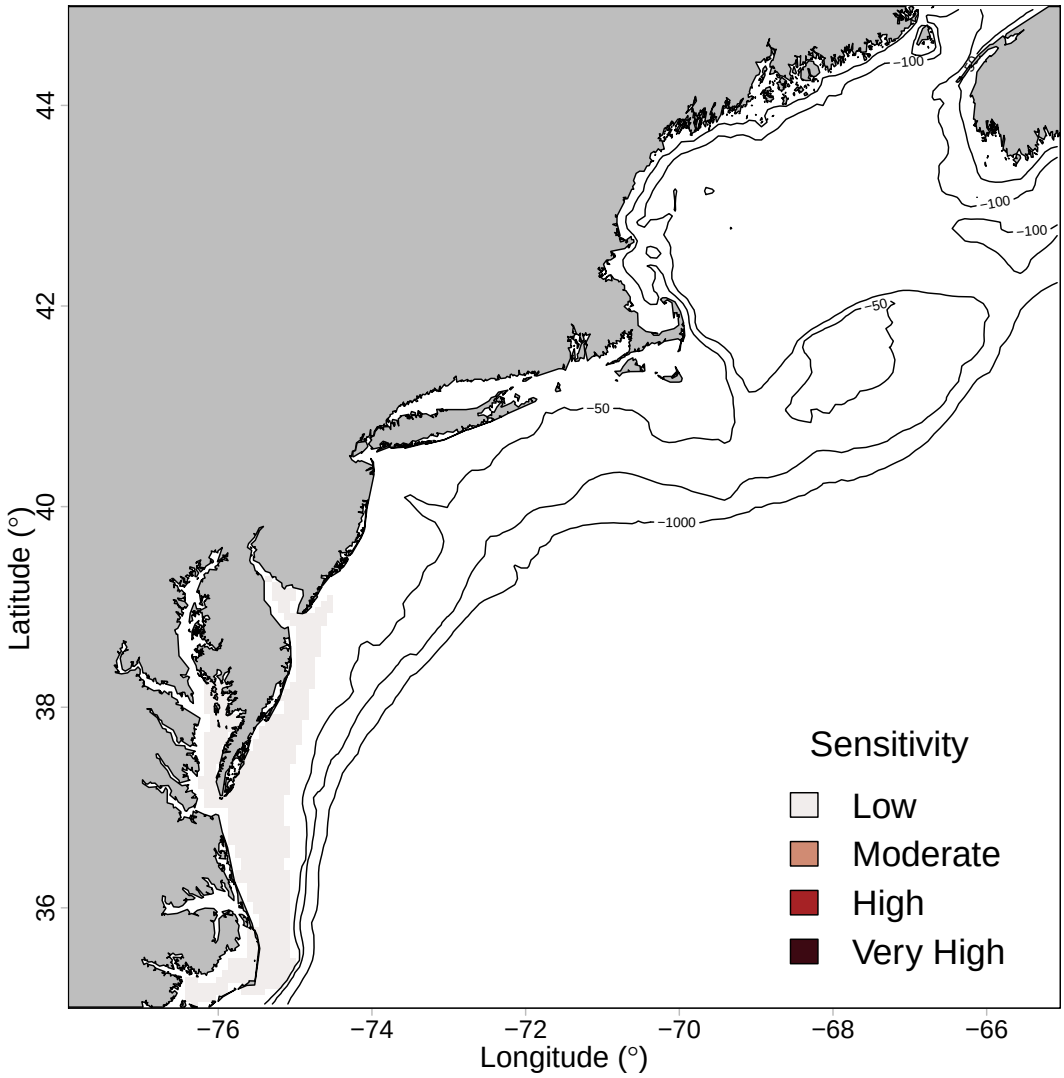


Figure 2: Cobia Sensitivity on the Northeast Shelf

Cobia received a **Low** sensitivity score. Certainty was 53% (Figure 1). The only attribute that received a score greater than 2.5 was Relationship to Calcified Organisms (Figure 1). Cobia primarily consume crustaceans, which make them vulnerable to ocean acidification (Smith 1995).

Cobia was not assessed in the 2016 Climate Vulnerability Assessment for the Northeast Shelf, but was assessed in the recent assessment for the Southeast Shelf. In this assessment, it received a Moderate sensitivity score. This score was driven by Dispersal of Early Life Stages, Early Life History Survival and Settlement Requirements, Reproductive Complexity, Spawning Cycle, and Sensitivity to Ocean Acidification (Burton et al. 2023).

Exposure:

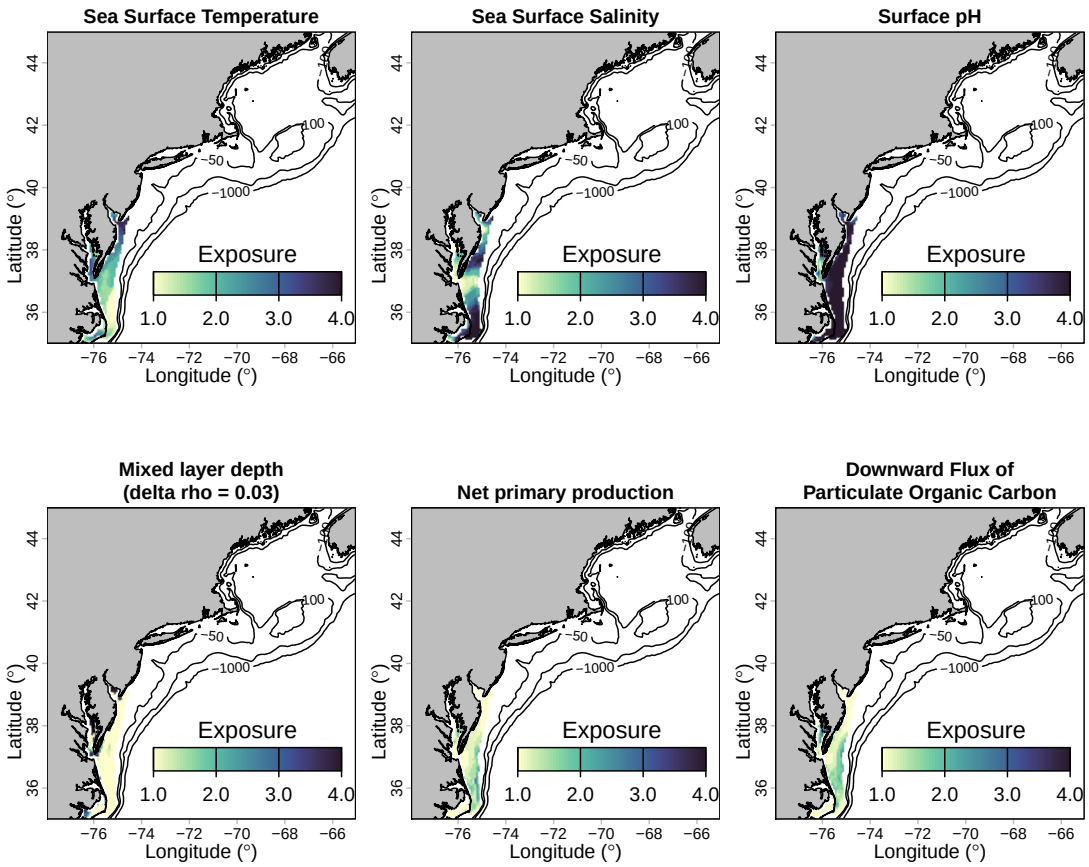


Figure 3: Maps of each Exposure Factor for Cobia. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure within Cobia's range on the Northeast Shelf was Moderate (2.17 +/- 0.91) (Figure 1). Exposure was greatest along the coasts of Delaware, Maryland, North Carolina, and in the mouth of the Delaware Bay. They were lowest at the mouth of the Chesapeake Bay (Figure 4). These patterns were driven by high surface salinity and pH exposures (Figure 3).

Cobia was not assessed in the 2016 Climate Vulnerability Assessment for the Northeast Shelf, but was assessed in the recent assessment for the Southeast Shelf. In this assessment, Cobia received a Very High exposure score, driven by sea surface temperature, salinity and ocean acidification (Burton et al. 2023).

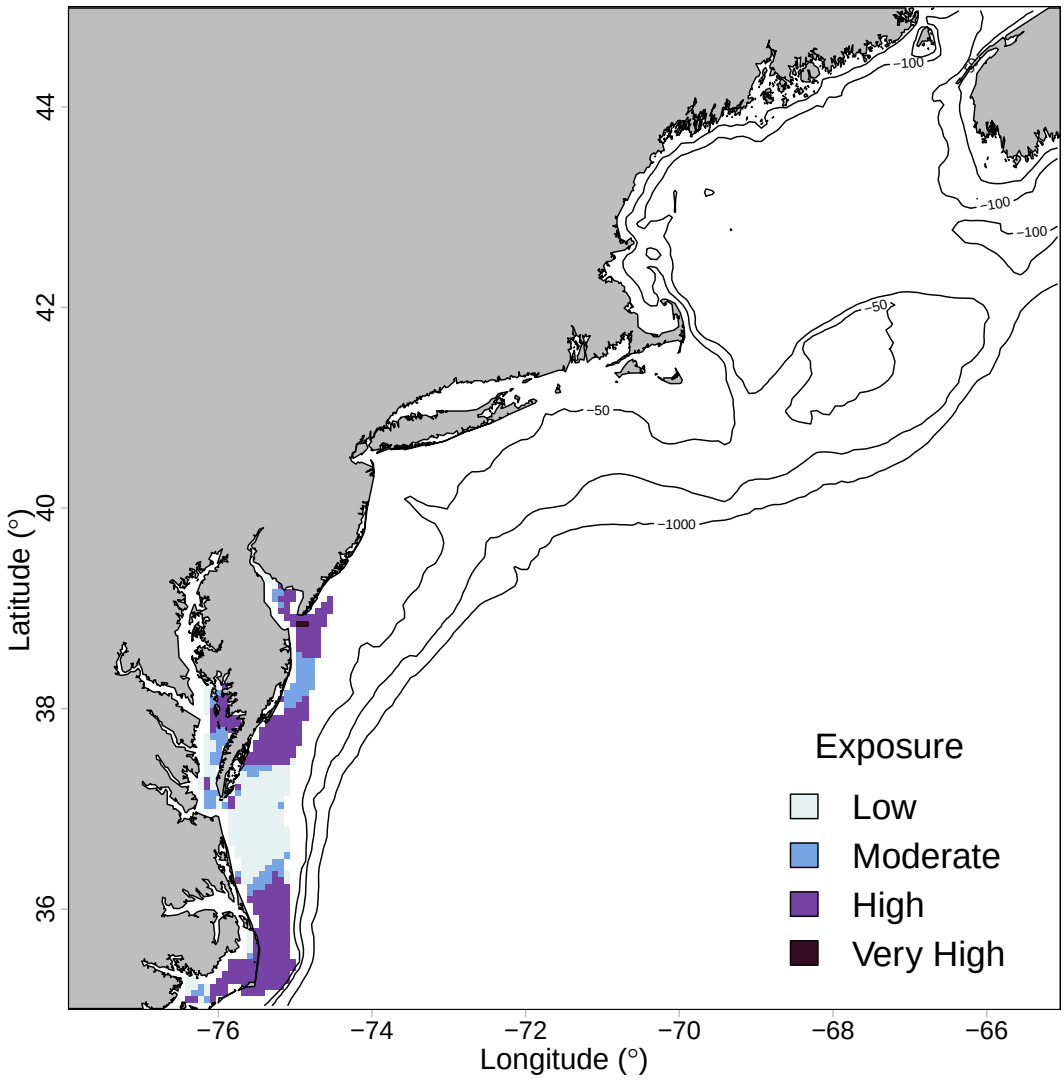


Figure 4: Total Exposure on the Northeast Shelf for Cobia. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

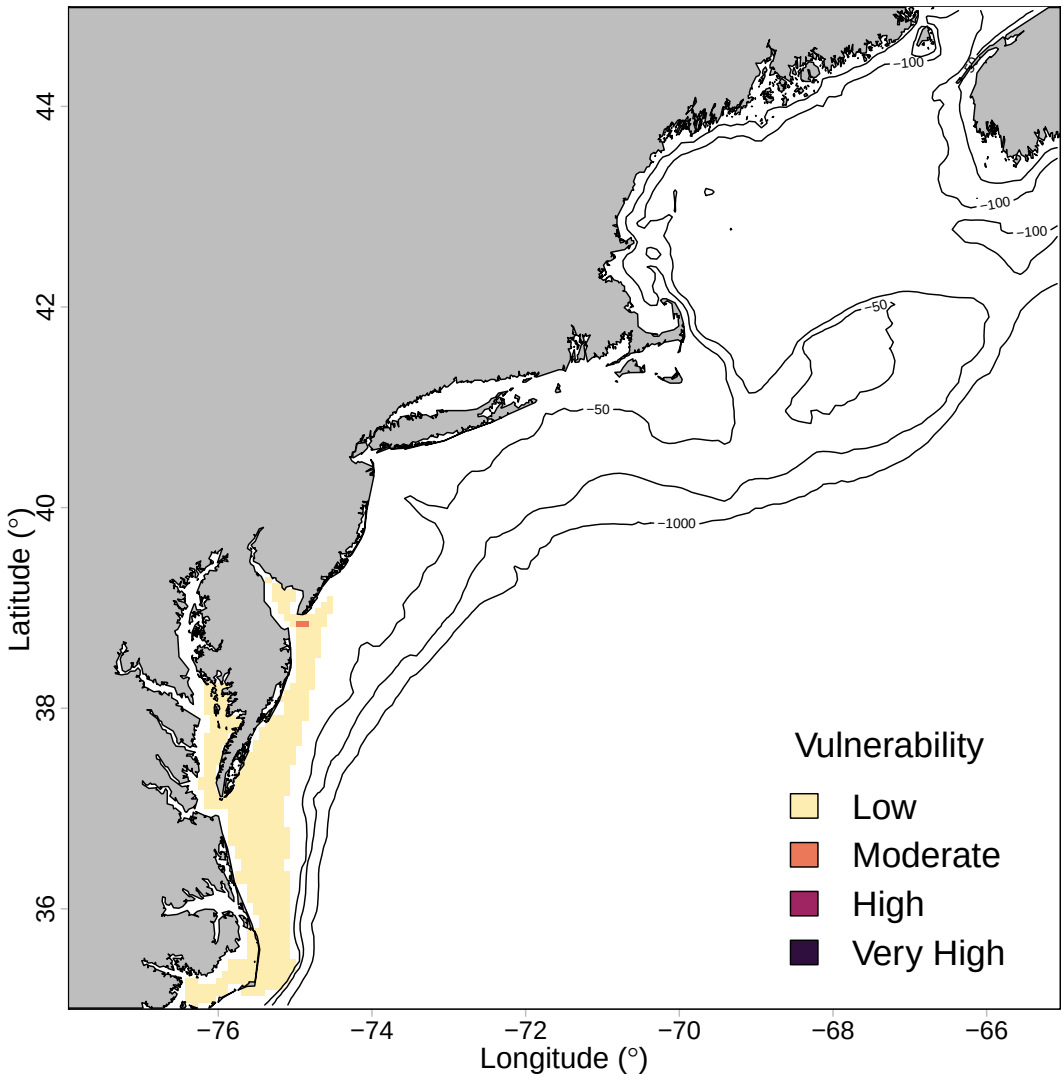


Figure 5: Vulnerability on the Northeast Shelf for Cobia. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability within Cobia's range on the Northeast Shelf was **Low** (2.17 +/- 0.91) (Figure 1). This was consistent across the species range in the region (Figure 5).

Cobia were given a High Vulnerability score in the recent assessment on the Southeast Shelf (Burton et al. 2023).

Data Quality Metrics

Life History Data Quality:

88% of data quality scores were greater than or equal to 2, indicating strong data quality. Egg and Larval Survival and Settlement Requirements and Juvenile Habitat Requirements were identified as data gaps

(mean attribute data quality score < 2) (Figure 1).

Model Confidence Score:

Mean model confidence for the ensemble Cobia model was 1.75 (+/- 0.5). Experts stated that broad trends were reasonable but a low number of presences (<100) decreased confidence. The model AUC was 0.98 indicating excellent predictive performance.

Additional Metrics

Distribution Shifts:

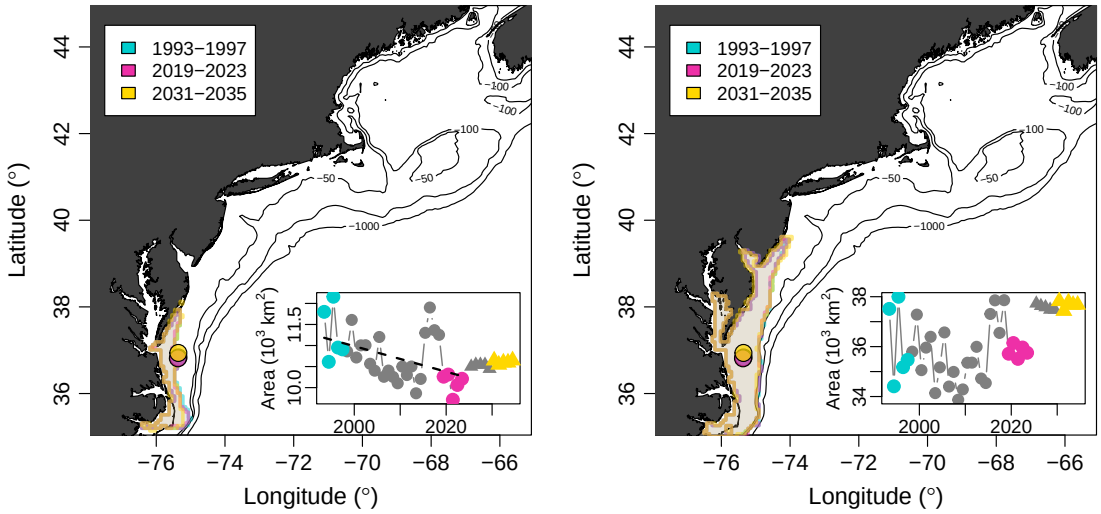


Figure 6: Distribution shifts on the Northeast Shelf for Cobia. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Cobia experienced a significant decline in core habitat area between 1993 and 2023. This was due to a loss of habitat on the continental shelf (Figure 6). The distribution centroid shifted 3 km southwest during this time period.

Under near future climate conditions, cobia habitat is not predicted to change significantly. The ensemble model predicted further habitat loss on the continental shelf, but ranges may continue to expand northward to compensate for the loss (Figure 6). The distribution centroid is predicted to shift 14 km northward.

In the recent Climate Vulnerability Assessment for the Southeast Shelf, cobia received a High Distributional Vulnerability Rank due to their high mobility and dispersive early life stages (Burton et al. 2023).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	11,355	36,349	3.80	1.52
2019-2023	2.5 km SW	10,160 (-1,194)	35,856 (-493)	3.80 (0)	1.44 (-0.08)
2031-2035	13.7 km N	10,595 (+435)	37,674 (+1,818)	3.87 (+0.06)	1.60 (+0.16)

Directional Effect:

Experts predicted that climate change would have a **neutral** effect on Cobia. Certainty was 95%. Warmer temperatures could lead to northward shifts in distributions and earlier seasonal migrations (Crear et al. 2020). The recent Southeast US Climate Vulnerability Assessment also predicted that climate change would have a neutral effect on Cobia (Burton et al. 2023).

Climate Effects on Abundance and Distribution

Cobia off the eastern U.S. Atlantic coast currently overwinter offshore in waters south of North Carolina. Suitable habitat increases in the summer months with warming temperatures. Habitat distribution models suggest increased habitat suitability northward, shifting distributions poleward, and possible earlier seasonal migrations (Crear et al. 2020). This shift northward could prompt revision of management regulations as the fishery expands to new states or regions. Early life stage cobia may be somewhat resistant to projected rises in CO^2 levels (Ebanks 2013), although juvenile cobia in a laboratory study exhibited poor growth performance and reduced feeding at elevated temperatures (Le et al. 2024).

Life History Synopsis

Cobia are a pelagic, continental shelf fish distributed from Massachusetts to Argentina but are most abundant from Virginia to the Gulf of America (Shaffer and Nakamura 1989). Two stocks are recognized in the United States: one along the Atlantic coast south to Georgia and the other along the Atlantic coast of Florida into the Gulf (SouthEast Data, Assessment, and Review (SEDAR) 2020). Cobia are batch spawners, and move inshore to spawn when temperatures reach 20-25°C. They spawn over a period of 4-6 weeks, peaking in June-July in Virginia (Joseph, Norcross, and Massmann 1964). Cobia eggs hatch within 36 hours of fertilization. Given the short time frame between fertilization and hatching, it is likely that eggs and larvae do not travel far from spawning grounds (Shaffer and Nakamura 1989). Little is known about the feeding habits of larval and juvenile cobia. Adult cobia are generalist carnivores, feeding on benthic-epibenthic crustaceans (*Portunids* and *Callinectes* sp.), teleosts, cephalopods, and elasmobranchs (Smith 1995). Juveniles are pelagic and often associated with structure for feeding and shelter. Juveniles and adults can be found near shipwrecks, buoys, shellfish beds, and submerged aquatic vegetation (SouthEast Data, Assessment, and Review (SEDAR) 2020). Few predators of cobia in U.S. Atlantic waters are known, although dolphinfish (*Coryphaena hippurus*) have been reported to prey on small cobia (Shaffer and Nakamura 1989). Over 50% of cobia are mature by age 2, and can live up to 15 years (SouthEast Data, Assessment, and Review (SEDAR) 2020). Adults are highly migratory. Cobia exhibit a general migratory trend of overwintering in south Florida and the Florida Keys waters, and

then migrating north along the Atlantic and Gulf coasts in late winter-early spring. They likely return to south Florida coastal waters in the late fall (Hendon, Franks, and Fulford 2008). The Atlantic migratory cobia are managed by the Atlantic States Marine Fisheries Commission. The population is not overfished and overfishing is not occurring as of the last benchmark stock assessment in 2020 (SouthEast Data, Assessment, and Review (SEDAR) 2020).

Supplemental Scores

Exposure Timeseries

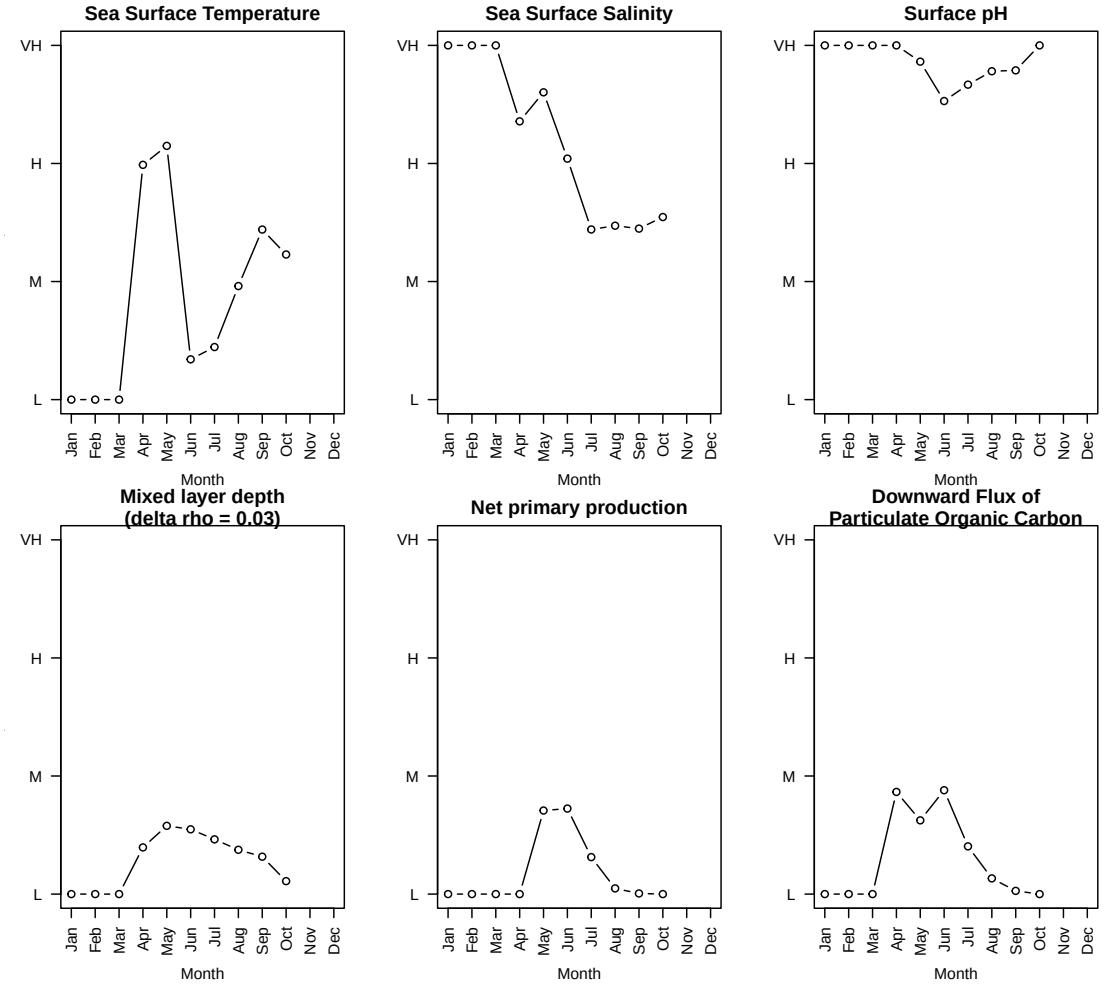


Figure 7: Monthly Timeseries of each Exposure Factor for Cobia. Exposure was calculated between 2014-2023 and 2025-2035

Exposure was High from January through June, due to high salinity and surface pH exposure. Exposures declined to Moderate to Low in the latter half of the year due to declines in temperature and salinity exposure (Figure 8; Figure 7).

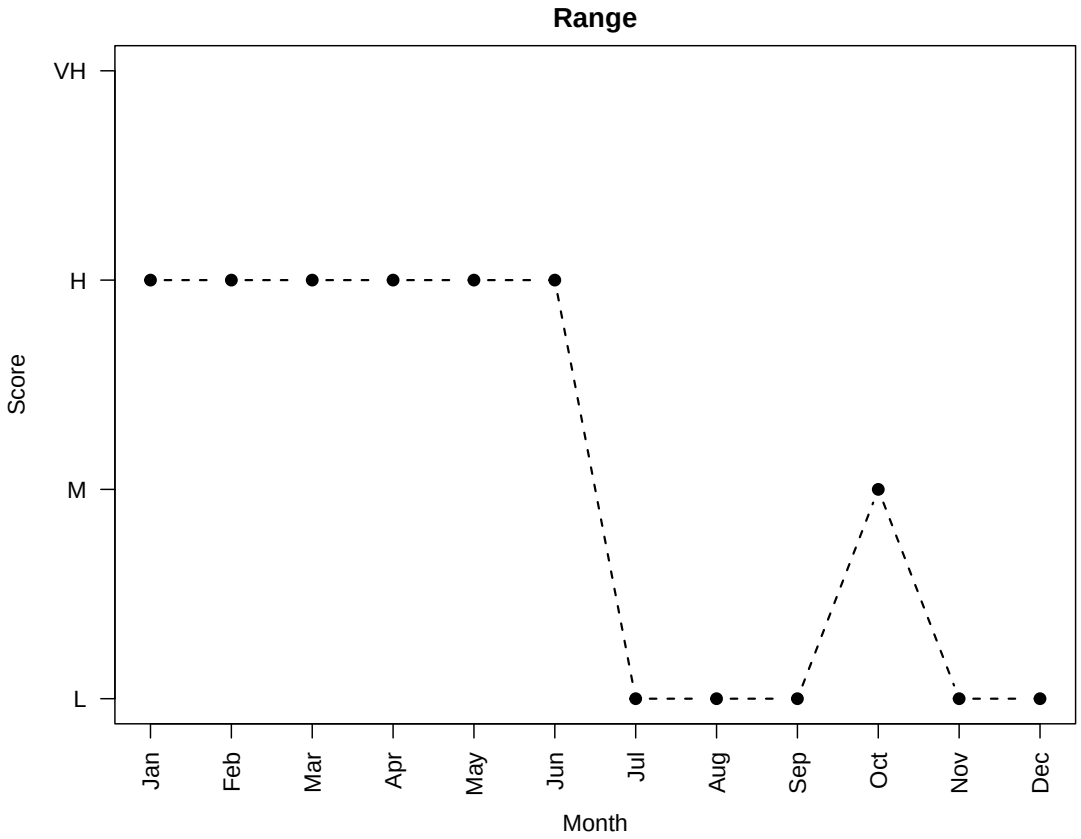


Figure 8: Timeseries of Total Exposure for Cobia

Exposure due to Significant Factors

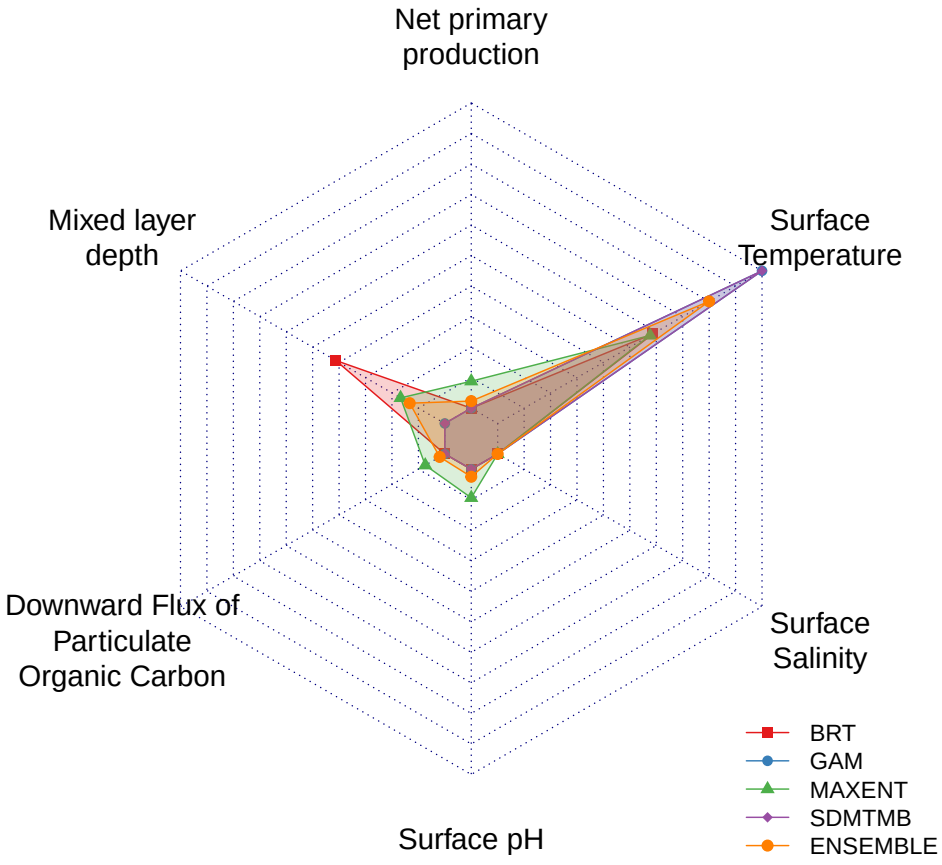


Figure 9: Dynamic Factors driving Cobia distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Cobia distributions in the region were driven by sea surface temperature and mixed layer depth (Figure 9). Cobia prefer warmer waters (Crear et al. 2020). These two variables did not have high exposures within the species range, and as a result, exposure was Low (1.05 +/- 0.25) when exposure was calculated only with these factors (Figure 10; Figure 11).

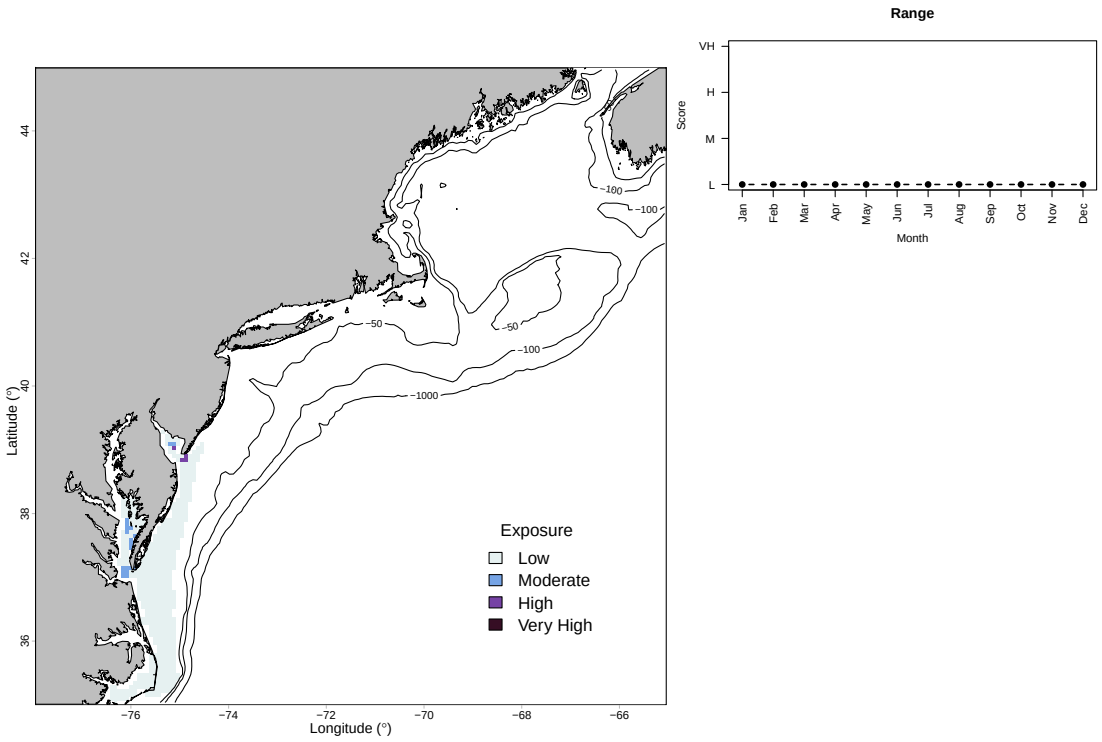


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Cobia distributions

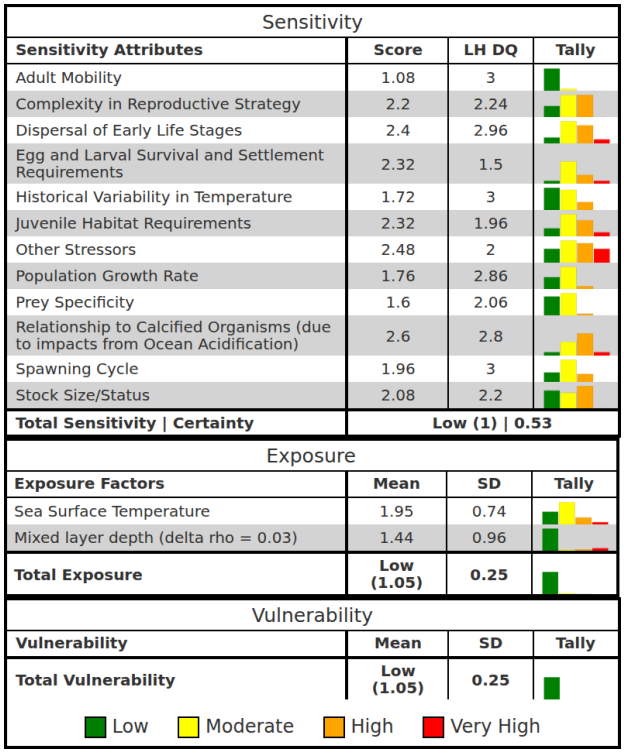


Figure 11: As in Figure 1 for Cobia, but with only significant Exposure factors and the resulting Vulnerability Scores.

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